

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI - K K BIRLA GOA CAMPUS

Semester-I (2025/2026)

Analog & Digital VLSI Design (EEE/INSTR F313)

Quiz-II (PART A)

Date : 28.10.2025

Duration : 30 Minutes

Maximum Marks : 15

Name of the student:

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ID. No.

2023A8PS08146

Question	Marks	Score
1	9	
2	6	
Total:	15	15

Question 1

Design of a CMOS inverter circuit:

Use the parameters given below:

nMOS : $V_{T0,n}=0.6\text{ V}$, $\mu_n C_{ox}=60\text{ }\mu\text{A/V}^2$

pMOS : $V_{T0,p}=-0.7\text{ V}$, $\mu_p C_{ox}=25\text{ }\mu\text{A/V}^2$

The power supply voltage is $V_{DD}=3.3\text{ V}$.

(a) Determine the (W/L) of PMOS if (W/L) of NMOS is 12 so that the switching (inversion) threshold voltage of the circuit is $V_{th}=1.3\text{ V}$. Remember V_{th} is switching threshold voltage of inverter.

(b) The CMOS fabrication process used to manufacture this inverter allows a variation of the $V_{T0,n}$ value by $\pm 10\%$ around its nominal value, and a variation of the $V_{T0,p}$ value by $\pm 15\%$ around its nominal value. Assuming that all other parameters (such as μ_n , μ_p , C_{ox} , $(W/L)_n$, $(W/L)_p$) always retain their nominal values, find the upper and lower limits of the switching threshold voltage (V_{th}) of this circuit.

$\frac{W}{L}$ of PMOS	8.35
$V_{th_{max}}$	1.375V
$V_{th_{min}}$	1.225V

Question 2

Consider symmetric CMOS inverter

Use the parameters given below:

nMOS : $V_{T0,n}=0.6\text{ V}$, $\mu_n C_{ox}=60\text{ }\mu\text{A/V}^2$

pMOS : $V_{T0,p}=-0.6\text{ V}$, $\mu_p C_{ox}=30\text{ }\mu\text{A/V}^2$

The power supply voltage is $V_{DD}=3.3\text{ V}$.

(a) Determine the (W/L) of PMOS if (W/L) of NMOS is 12.

(b) Find the value of V_{IL} and V_{IH}

$\frac{W}{L}$ of PMOS	24
V_{IL}	1.3875V
V_{IH}	1.9125V